

# 30-Day SQL Plan

Baraa's SQL Full Course (30 hrs) — 1 hour a day

## Your study window

You finish work at 4:00 AM and get home around 5:15 AM. Suggested slot: **5:15–6:15 AM**, right when you get home, before winding down. If that's too tired a time, an alternative is 1 hour in the late afternoon before your shift — pick whichever you can actually stick to, and keep it the same time daily.

## How to use this plan

- Code along in a real database (MySQL Workbench / SSMS / DB Fiddle) — never just watch.
- Track by topic completed, not the clock — if a lecture runs long, stop at a natural break and continue it tomorrow rather than rushing.
- Last 5–10 min of each session: write 2–3 things you learned, in your own words.
- Missed a day? Shift everything forward by one day — don't double up.
- Check the box when a day is done.

## BEGINNER



### 1 Day 1 — Installation & Setup

- Install MySQL / SSMS, connect to a database
- Tour of the tool interface

*Practice: Get your environment fully working before Day 2.*



### 2 Day 2 — SELECT Basics

- SELECT syntax fundamentals
- Selecting and aliasing columns

*Practice: Write 5 basic SELECT queries on a sample table.*



### 3 Day 3 — DDL Commands

- CREATE, ALTER, DROP
- Data types basics

*Practice: Create a small table from scratch, then alter it.*



### 4 Day 4 — DML Commands

- INSERT, UPDATE, DELETE

*Practice: Insert 5 rows, update 2, delete 1 — on your own table.*



### 5 Day 5 — Review + Rebuild

- Recap Beginner section

*Practice: Rebuild a small table from scratch, no notes.*

## INTERMEDIATE



## 6 Day 6 — Filtering Basics

- WHERE clause
- Comparison operators (=, <>, >, <, >=, <=)
- Logical operators: AND, OR, NOT

*Practice: Filter a table using 2–3 combined conditions.*

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## 7 Day 7 — Filtering Continued

- LIKE (wildcards % and \_)
- IN, NOT IN
- BETWEEN

*Practice: Write a query combining LIKE + BETWEEN on the same table.*

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## 8 Day 8 — Joins Part 1

- INNER JOIN
- LEFT JOIN / RIGHT JOIN

*Practice: Join 2 tables; explain in your own words how INNER vs LEFT differ.*

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## 9 Day 9 — Joins Part 2

- FULL (OUTER) JOIN
- Self joins
- Cross joins

*Practice: Write a self-join (e.g. employees-to-manager).*

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## 10 Day 10 — Set Operators

- UNION vs UNION ALL
- INTERSECT
- EXCEPT / MINUS

*Practice: Combine 2 queries with UNION, then UNION ALL — compare row counts.*

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## 11 Day 11 — String Functions

- CONCAT, SUBSTRING, TRIM
- UPPER/LOWER, LENGTH, REPLACE

*Practice: Clean up messy text data using 3+ chained string functions.*

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## 12 Day 12 — Numeric Functions

- ROUND, CEILING, FLOOR
- ABS, MOD, POWER

*Practice: Format a numeric column for reporting (round to 2 decimals).*

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## 13 Day 13 — Date Functions

- GETDATE/NOW, DATEADD, DATEDIFF
- Date formatting

*Practice: Calculate age or days-between from a date column.*

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## 14 Day 14 — NULL Handling

- IS NULL / IS NOT NULL
- COALESCE, ISNULL/IFNULL

*Practice: Write a query that replaces NULLs with a default value.*

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## 15 Day 15 — Aggregate Functions + Review

- COUNT, SUM, AVG, MIN, MAX
- GROUP BY, HAVING

*Practice: No notes: 1 query with a join, 1 with GROUP BY/HAVING, 1 with a string or date function.*

# WINDOW FUNCTIONS (extra focus)



## 16 Day 16 — Window Function Basics

- OVER()
- PARTITION BY

*Practice: Write a query with OVER() and PARTITION BY on a sample dataset.*



## 17 Day 17 — Ranking Functions

- ROW\_NUMBER
- RANK, DENSE\_RANK

*Practice: Rank rows within groups using all three functions; compare outputs.*



## 18 Day 18 — Aggregate Window Functions

- Running totals
- Aggregate functions with OVER()

*Practice: Build a running total column for a sales-style table.*



## 19 Day 19 — Window Functions Practice

- Consolidation — no new material

*Practice: Write 4–5 window function queries unaided, from memory.*

# ADVANCED



## 20 Day 20 — Subqueries

- Subqueries in SELECT, WHERE, FROM

*Practice: Write one query with each type of subquery placement.*



## 21 Day 21 — CTEs

- WITH clause
- Recursive CTEs (intro)

*Practice: Rewrite one of your subqueries as a CTE instead.*



## 22 Day 22 — Views

- CREATE VIEW
- When and why to use views

*Practice: Create a view summarizing a table, then query it.*



## 23 Day 23 — CTAS & Temp Tables

- CREATE TABLE AS SELECT
- Temporary tables

*Practice: Create a temp table from a query and use it in a second query.*

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- ☐ 24 **Day 24 — Stored Procedures**
  - CREATE PROCEDURE
  - Parameters*Practice: Write a simple stored procedure that filters by a parameter.*
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- ☐ 25 **Day 25 — Triggers**
  - CREATE TRIGGER
  - BEFORE/AFTER events*Practice: Write a trigger that logs changes to a table.*
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- ☐ 26 **Day 26 — Indexes, Partitions & Optimization**
  - Indexes basics
  - Partitions
  - Key optimization tips*Practice: Identify 2 places an index would help in a query you've written.*
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- ☐ 27 **Day 27 — AI Integration in SQL**
  - Using AI tools alongside SQL workflows*Practice: Try using AI to explain or optimize one of your earlier queries.*
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## HANDS-ON PROJECTS

- ☐ 28 **Day 28 — Project: Data Warehouse**
  - Hands-on build*Practice: Follow along and build; expect this to run over 1 hour — split across sessions if needed.*
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- ☐ 29 **Day 29 — Project: EDA**
  - Exploratory Data Analysis project*Practice: Apply joins, aggregates, and window functions together.*
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- ☐ 30 **Day 30 — Project: Advanced Analytics / AI + Final Review**
  - Advanced Analytics/AI-driven project
  - Or: finish any lagging project*Practice: Without notes, write one query each using: a join, a subquery/CTE, and a window function.*
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**Final tip:** Days 28–30 are hands-on projects, not lecture-watching — they may take longer than 1 hour. That's expected; don't rush the projects, they're where everything ties together.